MATH40003 - Linear Algebra and Groups

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Introduction

Why do we study linear algebra?

- 2D and 3D Geometry
- Signal processing: audio and video samples, data and communications
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Chapter 1

Fields and Vector Spaces

1.1 Fields

Definition 1.1 (Field)**.** Let F be a set with two binary operations, addition $+ : F \times F \to F$ and multiplication $\cdot : F \times F \to F$. We say that F is a *field* if it satisfies the following axioms for all $a, b, c \in F$:
Axioms for Addition $(F, +)$**:**

A1. Associativity: $(a + b) + c = a + (b + c)$

A2. Commutativity: $a + b = b + a$

A3. Additive Identity: There exists an element $0 \in F$ such that $a + 0 = a$.

A4. Additive Inverse: For each $a \in F$, there exists an element $-a \in F$ such that $a + (-a) = 0$.

Axioms for Multiplication $(F, \cdot)$**:**

M1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

M2. Commutativity: $a \cdot b = b \cdot a$

M3. Multiplicative Identity: There exists an element $1 \in F$, with $1 \neq 0$, such that $a \cdot 1 = a$.

M4. Multiplicative Inverse: For each $a \in F \setminus \{0\}$ there exists an element $a^{-1} \in F \setminus \{0\}$ such that $a \cdot a^{-1} = 1$.

Distributivity:

D1. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Remark. This is all the structure we need from the real numbers to do linear algebra. We can do linear algebra with many other fields, not just $\mathbb{R}$, like the finite field $\mathbb{F}_2$. Abstracting the scalars into an arbitrary field F allows for some rich and interesting results in linear algebra.

Example 1.2. The integers modulo p, $\mathbb{F}_p \coloneqq \{0, 1, \ldots, p - 1\}$, where p is prime, form a field. Addition and multiplication are defined modulo p. All properties except the existence of a multiplicative inverse follow easily from the fact that $\mathbb{Z}$ is a commutative ring.

Lemma 1.3. *In $\mathbb{F}_p$, $xy = 0 \iff x = 0$ or $y = 0$.*

Proof. $xy = 0 \implies p \mid xy \implies p \mid x$ or $p \mid y$ (by Euclid's lemma). This means $x \equiv 0 \pmod p$ or $y \equiv 0 \pmod p$. □

Theorem 1.4. *For every $a \in \mathbb{F}_p \setminus \{0\}$ there exists $a^{-1} \in \mathbb{F}_p \setminus \{0\}$ such that $aa^{-1} = 1 = a^{-1}a$.*

Proof. Given $a \in \mathbb{F}_p$, by commutativity we only need to find a $b \in \mathbb{F}_p$ such that $ab = 1$. Finding an explicit construction for b is difficult, so we instead show that the set of multiples $S := \{a \cdot 0, a \cdot 1, \dots, a \cdot (p-1)\}$ is exactly equal to $\mathbb{F}_p$.

It suffices to show that the p elements $a \cdot 0, a \cdot 1, \dots, a \cdot (p-1) \in \mathbb{F}_p$ are all distinct. Suppose $a \cdot c = a \cdot d$. Then $a \cdot (c - d) = 0$.

By our lemma, since $a \neq 0$ ($p \nmid a$), we must have $c - d = 0$, which implies $c = d$. Thus, multiplication by a maps distinct elements to distinct elements. Since there are p elements in $\mathbb{F}_p$, the set S covers all of $\mathbb{F}_p$. Because $1 \in \mathbb{F}_p$, there must be some element $b \in \{0, \dots, p-1\}$ such that $ab = 1$. □

Corollary 1.5. *$\mathbb{F}_p$ is a field.*

Remark. The proof above relies on the Pigeonhole Principle. Because multiplication by a non-zero element a is injective on the finite set $\mathbb{F}_p$, it must also be surjective.

Definition 1.6 (Subfield)**.** A subset $F \subseteq E$ of a field E is a *subfield* if $1 \in F$ and it is closed under arithmetic: whenever $a, b \in F$ we have $a + b, ab \in F$; if $a \in F$ then $-a \in F$; and if $a \in F \setminus \{0\}$ then $a^{-1} \in F$.

Remark. This definition is slightly redundant. To prove a subset $F \subseteq E$ is a subfield, it actually suffices to show closure under addition and multiplication, the existence of multiplicative inverses for non-zero elements, and that $-1 \in F$. Because if $-1 \in F$ and we have the closures above, then $(-1)(-1) = 1 \in F$ and $(-1) + 1 = 0 \in F$. Additive inverses follow from closure of addition with -1.

Example 1.7. Some examples of fields and subfields are:

- $\mathbb{Q}[i] := \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field (the "Gaussian rationals"). It is a subfield of $\mathbb{C}$.
- $\mathbb{Q}[\sqrt{2}] := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$ is a field, and it is a subfield of $\mathbb{R}$ (exercise: show this).
- The field of rational functions $F(X) := \left\{\frac{p(X)}{q(X)} : p, q \in F[X],\ q \neq 0\right\}$ is a field. $\mathbb{Q}(x)$ is a subfield of $\mathbb{R}(x)$, for example.

1.2 Vector Spaces

Definition 1.8. The vector space F^n of n-tuples over a field F is defined as:

$$F^n := \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_1, \ldots, x_n \in F \right\}$$

Remark. This is the primordial example of a vector space. Every finite-dimensional vector space is structurally identical (isomorphic) to F^n. The geometric intuition of $\mathbb{R}^n$ (arrows, scaling, and adding tip-to-tail) carries over algebraically to arbitrary fields.

Definition 1.9 (Commutative Group)**.** A commutative group $(V, +)$ is a set equipped with a binary operation $+ : V \times V \to V$ satisfying:

- **Associativity:** For all $u, v, w \in V$, $u + (v + w) = (u + v) + w$.
- **Commutativity:** For all $v, w \in V$, $v + w = w + v$.
- **Additive Identity:** There exists $0 := (0, \ldots, 0) \in V$ such that for all $v \in V$, $v + 0 = 0 + v = v$.
- **Additive Inverse:** For all $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0 = (-v) + v$.

Lemma 1.10. *In a commutative group $(V, +)$, the zero vector and additive inverses are unique.*

Proof. Given another zero vector $0'$, we have $0' = 0 + 0' = 0$. Thus, the zero vector is unique.

If w also acts as an additive inverse to v (so $w + v = 0$), then:

$$w = w + 0 = w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v.$$

Thus, the additive inverse is unique. □

Definition 1.11 (Vector Space)**.** A vector space V over a field F is a commutative group $(V, +)$ together with an action $F \times V \to V$, mapped by $(\lambda, v) \mapsto \lambda v$, called *scalar multiplication*, satisfying the following properties for all $\lambda, \mu \in F$ and $v, w \in V$:

- **Associativity:** $\lambda(\mu v) = (\lambda\mu)v$
- **Distributivity of scalar sums:** $(\lambda + \mu)v = \lambda v + \mu v$
- **Distributivity of vector sums:** $\lambda(v + w) = \lambda v + \lambda w$
- **Unit multiplication:** $1 \cdot v = v$

Remark. Structurally, one can think of a vector space as having two tiers: the "ground" field F provides the scalars (stretching factors), while the commutative group V provides the vectors (the objects being stretched and added; indeed, F acts on V to get scalar multiples of vectors). The axioms ensure these tiers interact in a linear way.

Example 1.12. Other examples of vector spaces include:

- $F[X] = \{a_0 + a_1X + \cdots + a_nX^n : a_i \in F\}$ (polynomials).
- Subfields $F \subseteq E$ (E acts as a vector space over its subfield F).
- $\mathrm{Fun}(X, F) = \{f : X \to F\}$ (functions from an arbitrary set X into F).

Chapter 2

Matrices

2.1 Matrices and Linear Maps

Matrices are the "coordinated" way to describe linear transformations. We will formalise this in Theorem 2.3 below. But first, we define a linear transformation.

Definition 2.1 (Linear transformation)**.** Fix a field F and two vector spaces V, W over F. A *linear transformation* (or linear map) is a function $T : V \to W$ such that:

- **Additivity:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in F$ and $v \in V$.

Lemma 2.2. *If $T : V \to W$ is a linear transformation, then $T(0) = 0$.*

Proof. By linearity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding $-T(0)$ (the additive inverse of $T(0)$ in W) to both sides yields $T(0) = 0$. $\square$

Recall that any finite-dimensional vector space V, with dimension n, is isomorphic to F^n. For now, when we say we have written V and W in coordinates, we mean we replace V and W with F^n and F^m. This motivates the concept of matrices.

Theorem 2.3. *For all linear maps $T : F^n \to F^m$ defined by $(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m)$, there exists an array of numbers (a_{ij}) such that:*

$$y_i = \sum_{j=1}^{n} a_{ij} x_j$$

Proof. We define the standard elementary vectors $e_j \in F^n$ to have value 1 in the j-th entry, and 0 everywhere else. Since $T(e_j)$ is a vector in F^m, we can write it as a list of m coordinates. For all j, define $a_{1j}, \dots, a_{mj} \in F$ by:

$$T(e_j) = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

(Idea: think of $T(e_j)$ as the j^{th} column of the matrix (a_{ij})).

Any input vector can be written as $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$ by our definition of addition in F^n. By the linearity of T, we have:

$$\begin{aligned} T(x_1, \ldots, x_n) &= T(x_1 e_1 + \cdots + x_n e_n) \\ &= x_1 T(e_1) + \cdots + x_n T(e_n) \\ &= \sum_{j=1}^{n} x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} \end{aligned}$$

Taking the i^{th} row (coordinate) of the above sum, we get:

$$y_i = \sum_{j=1}^{n} a_{ij}\, x_j$$

as required. □

Remark. The converse is also true: any function $T : F^n \to F^m$ given by such an array (a_{ij}) is a linear map. This is fundamental: *linear maps are exactly the functions that can be represented by matrices.*

Observe that all the information needed to compute this linear transformation is encoded in the array $A \coloneqq (a_{ij})$.

Definition 2.4 (Matrix)**.** The set of $m \times n$ matrices over a field F is defined as:

$$M_{m\times n}(F) \coloneqq \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}$$

How do we conveniently encode the action of T using this? We define matrix-vector multiplication:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The upshot is that, given a linear map $T : F^n \to F^m$, we *define* matrix multiplication with a column vector precisely to execute this map.

2.2 Matrix Multiplication

2.2.1 Motivation

First, we motivate the definition of general matrix multiplication. Consider two linear maps:

- $T : F^k \to F^n$ represented by the matrix $B \in M_{n\times k}(F)$
- $S : F^n \to F^m$ represented by the matrix $A \in M_{m\times n}(F)$

Define the composite linear map $R : F^k \to F^m$ by $R = S \circ T$. By Claim ($\star$), this new linear map must also have a matrix representation, which we will call $C \in M_{m\times k}(F)$.

We *want* to calculate $c_{i\ell}$, the $(i,\ell)^{\text{th}}$ element of the matrix C. Let's calculate the ℓ^{th} column of C, call it C_ℓ. We know that each column of C represents the effect of the linear map R on the standard basis vector e_ℓ:

$$C_\ell = R(e_\ell) = S(T(e_\ell))$$

We know $T(e_\ell)$ is exactly the ℓ^{th} column of B, call it B_ℓ. So we must calculate $S(B_\ell)$. Because B_ℓ is a column vector in F^n, evaluating $S(B_\ell)$ is equivalent to taking a linear combination of the columns of A (which we will denote $A_1, \dots, A_n$):

$$S(B_\ell) = b_{1\ell}A_1 + \cdots + b_{n\ell}A_n$$

To find $c_{i\ell}$, we take the i^{th} row of this result:

$$\begin{aligned} c_{i\ell} = (C_\ell)_i &= b_{1\ell}(A_1)_i + \cdots + b_{n\ell}(A_n)_i \\ &= b_{1\ell}a_{i1} + \cdots + b_{n\ell}a_{in} \\ &= \sum_{j=1}^{n} a_{ij}b_{j\ell} \end{aligned}$$

This is exactly the formula for $c_{i\ell}$ that we use to define matrix multiplication. Therefore, the fact that composition of linear maps gives the formula above for the numbers $c_{i\ell}$ is why we define matrix multiplicaation in the following way:

Definition 2.5 (Matrix operations)**.**

- Matrix multiplication is the function $\cdot : M_{m\times n}(F) \times M_{n\times p}(F) \to M_{m\times p}(F)$ defined by:

$$(a_{ij})_{m\times n}\,(b_{k\ell})_{n\times p} := \left(c_{i\ell} = \sum_{j=1}^{n} a_{ij}b_{j\ell} \right)_{m\times p}$$

- Matrix addition is the function $+ : M_{m\times n}(F) \times M_{m\times n}(F) \to M_{m\times n}(F)$ defined by:
$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

- Scalar multiplication for matrices is the function $\cdot : F \times M_{m\times n}(F) \to M_{m\times n}(F)$ defined by:
$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Remark. Just like $F^n \cong M_{n\times 1}(F)$, the set $M_{m\times n}(F)$ forms a vector space under matrix addition and scalar multiplication. However, matrix multiplication gives $M_{n\times n}(F)$ *additional* structure beyond just a vector space (it forms an algebra).

Definition 2.6 (Square Matrix)**.** A matrix is square if $A \in M_{n\times n}(F)$ (the number of rows equals the number of columns). Square matrices of a fixed size can be multiplied by each other.

Proposition 2.7. *Matrix multiplication is associative:* $(AB)C = A(BC)$.

Proof.

$$((AB)C)_{i\ell} = \sum_j (AB)_{ij} c_{j\ell} = \sum_j \left(\sum_k a_{ik} b_{kj} \right) c_{j\ell} = \sum_{j,k} a_{ik} b_{kj} c_{j\ell}$$

$$(A(BC))_{i\ell} = \sum_k a_{ik} (BC)_{k\ell} = \sum_k a_{ik} \left(\sum_j b_{kj} c_{j\ell} \right) = \sum_{j,k} a_{ik} b_{kj} c_{j\ell}$$

Therefore, $((AB)C)_{i\ell} = (A(BC))_{i\ell}$. □

Corollary 2.8. *Given linear maps* $S : F^m \to F^n$ *and* $T : F^n \to F^p$ *with matrix representations* A *and* B *(so* $S(v) = Av$ *and* $T(v) = Bv$*), the matrix representation of the composite map* $T \circ S : F^m \to F^p$ *is given by the matrix product* BA.

Proof. $T \circ S(v) = T(S(v)) = B(Av) = (BA)v$ by associativity. □

2.2.2 Some more formal properties of matrices

Definition 2.9 (Bilinear map)**.** Let V, W, U be vector spaces over F. A map $f : V \times W \to U$ mapping $(v, w) \mapsto f(v, w)$ is *bilinear* if it is linear in each argument when the other is held fixed:

- $(\lambda_1 v_1 + \lambda_2 v_2, w) \mapsto \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$
- $(v, \lambda_1 w_1 + \lambda_2 w_2) \mapsto \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$

Remark. Note that this implies the standard scalar extraction property $f(\lambda v, w) = f(v, \lambda w) = \lambda f(v, w)$ (simply set $\lambda_2 = 0$).

Example 2.10. The dot product $F^n \times F^n \to F$, defined by $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto a_1 b_1 + \cdots + a_n b_n$ is bilinear. This follows easily from the distributive properties of the field F.

We want to prove that matrix multiplication is bilinear. To do this, we first establish how matrix multiplication interacts with columns and rows.

Proposition 2.11. *Let* $A \in M_{m\times n}(F)$ *and* $B = (B_1, \dots, B_p) \in M_{n\times p}(F)$, *where* B_j *are the columns of* B. *Then:*

$$AB = (AB_1, \dots, AB_p)$$

Proof. By the formula, $(AB)_{i\ell} = \sum_k a_{ik} b_{k\ell}$. Notice that $\sum_k a_{ik} b_{k\ell}$ is exactly the i^{th} element of the matrix-vector product AB_ℓ. Thus, the ℓ^{th} column of AB is exactly the vector AB_ℓ. □

Remark. This provides a *column-wise* way to calculate matrix products. To find the entire j^{th} column of AB, just calculate the matrix-vector product AB_j. The j^{th} column of AB **only** depends on the j^{th} column of B.

Proposition 2.12. *We can separate rows under right multiplication. Let B_i be the rows of B. Then:*

$$\begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix} A = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix}$$

Proof. Same logic as above, switching indices appropriately. □

Corollary 2.13. *Matrix multiplication is bilinear.*

Proof. We prove linearity in the second argument. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and $\lambda \in F$.

$$\begin{aligned} A(\lambda B) &= A(\lambda B_1 \dots \lambda B_n) \\ &= (A(\lambda B_1) \dots A(\lambda B_n)) && \text{(Column separation)} \\ &= (\lambda(AB_1) \dots \lambda(AB_n)) && \text{(Matrix-vector linearity)} \\ &= \lambda(AB_1 \dots AB_n) \\ &= \lambda(AB) \end{aligned}$$

And for addition:

$$\begin{aligned} A(B+C) &= A(B_1 + C_1 \dots B_n + C_n) \\ &= (A(B_1 + C_1) \dots A(B_n + C_n)) \\ &= (A(B_1) + A(C_1) \dots A(B_n) + A(C_n)) \\ &= (A(B_1) \dots A(B_n)) + (A(C_1) \dots A(C_n)) \\ &= AB + AC \end{aligned}$$

Linearity in the first argument follows similarly. □

Definition 2.14 (Transpose)**.** The transpose of a matrix $A = (a_{ij}) \in M_{m \times n}(F)$ is defined as $A^t = (a_{ji}) \in M_{n \times m}(F)$.

Proposition 2.15. *The transpose operation satisfies the following properties:*

- *Transpose is linear: $(A+B)^t = A^t + B^t$ and $(\lambda A)^t = \lambda A^t$.*
- $(A^t)^t = A$.
- $(AB)^t = B^t A^t$.

Proof. We prove the product rule. Let $C = AB$. Then $c_{ij} = \sum_k a_{ik} b_{kj}$. The (i,j) entry of $(AB)^t$ is the (j,i) entry of AB, which is $\sum_k a_{jk} b_{ki}$. The (i,j) entry of $B^t A^t$ is the dot product of the i^{th} row of B^t and the j^{th} column of A^t. This is exactly $\sum_k (B^t)_{ik} (A^t)_{kj} = \sum_k b_{ki} a_{jk} = \sum_k a_{jk} b_{ki}$. □

Remark. Notice that the order reverses in $(AB)^t = B^t A^t$. This is necessary for the dimensions to align. For example, if A is 1×5 and B is 5×1, AB is 1×1. A^t is 5×1 and B^t is 1×5. The product $A^t B^t$ is impossible, but $B^t A^t$ results in a 1×1 matrix. There is a deeper reason for this, which we will see when we look at dual vector spaces and dual linear transformations.

2.2.3 Special Matrices

Definition 2.16 (Symmetric Matrix)**.** A square matrix A is *symmetric* if $A = A^t$.

Definition 2.17 (Orthogonal Matrix)**.** A square matrix A is *orthogonal* if $AA^t = I$.

Remark. For a square matrix, $AA^t = I \iff A^t A = I$. What does it mean for $AA^t = I$? If A_i are the rows of A, then $(AA^t)_{ij} = A_i \cdot A_j$. Therefore:

$$A_i \cdot A_j = \delta_{ij} := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

This means the *rows* of A are orthonormal. Similarly, $A^t A = I$ implies the *columns* of A are orthonormal. Geometrically in $\mathbb{R}^n$, multiplying by an orthogonal matrix represents a rigid transformation (a rotation or reflection) that preserves lengths and angles.

Definition 2.18 (Triangular and Diagonal Matrices)**.** Let $A = (a_{ij})$ be a matrix.

- A is *upper-triangular* if $a_{ij} = 0$ for all $i > j$.
- A is *lower-triangular* if $a_{ij} = 0$ for all $i < j$.
- A is *diagonal* if it is both upper and lower triangular (i.e., $a_{ij} = 0$ for all $i \neq j$).

Chapter 3

Linear Equations

3.1 Solving Linear Systems

3.1.1 Gaussian Elimination

A system of m linear equations in n variables can be encoded by the matrix equation:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

Definition 3.1 (Row equivalence)**.** Let $A, B \in M_{m\times n}(F)$. We say that A and B are *row-equivalent* if B can be obtained from A by a finite sequence of the following elementary row operations:

- Permuting two rows.
- Multiplying a row by a non-zero scalar.
- Adding a scalar multiple of one row to another row.

Definition 3.2 (Row Echelon Form (REF))**.** A matrix is in *row echelon form* if:

- All rows consisting entirely of zeros are at the bottom.
- The leading non-zero entries (pivots) of each non-zero row are 1.
- The pivots of each non-zero row occurs strictly to the right of the pivot in the previous row.

Remark. Gaussian elimination is an algorithm that uses elementary row operations to transform any matrix into REF. This allows us to solve linear systems by back-substitution.

Example 3.3. Consider the augmented matrix of a linear system:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 5 & 3 & 10 \\ -1 & -1 & 1 & -1 \end{array}\right)$$

We apply Gaussian elimination to transform it into REF:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{array}\right) \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right) R_3 \to R_3 - R_2$$

This matrix is now in REF.

3.1.2 Gauss-Jordan Elimination

Definition 3.4 (Reduced Row Echelon Form (RREF)). A matrix is in *reduced row echelon form* if:

- It is in row echelon form.
- Every pivot entry is equal to 1.
- Each pivot is the only non-zero entry in its column.

Remark. Gauss-Jordan elimination continues the process of Gaussian elimination to transform a matrix from REF into RREF. In RREF, the pivot columns are exactly the standard basis vectors e_i. Solving a system in RREF requires no back-substitution; each pivot equation directly expresses a pivot variable in terms of the free variables.

Example 3.5. Continuing from the previous example, we apply Gauss-Jordan elimination to reach RREF:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right) \begin{array}{l} R_1 \to R_1 - R_3 \\ R_2 \to R_2 - R_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right) R_1 \to R_1 - 2R_2$$

The matrix is now in RREF. The solution is uniquely determined: $x_1 = 1, x_2 = 1, x_3 = 1$.

Theorem 3.6. *Every matrix is row-equivalent to a matrix in RREF.*

Proof. This is achieved constructively by the Gauss-Jordan elimination algorithm. □

3.1.3 Uniqueness of RREF

Theorem 3.7. *An RREF augmented matrix $(A|b)$ represents an inconsistent system if and only if the last column is a pivot column.*

Proof. If the last column is a pivot column, the matrix contains a row of the form $(0 \ \cdots \ 0 \mid 1)$, which encodes the contradictory equation $0 = 1$. Thus, the system is inconsistent.

Conversely, if the system is inconsistent, its RREF must contain a contradiction of the form $(0 \ \cdots \ 0 \mid c)$ for some $c \neq 0$. Scaling this row by c^{-1} leaves a 1 in the final column, making it a pivot column. □

Theorem 3.8 (Equivalence of Linear Systems)**.**

1. *If $(A|b)$ is row-equivalent to $(A'|b')$, then the systems $Ax = b$ and $A'x = b'$ have the same solution set.*
2. *Conversely, if $(A|b)$ and $(A'|b')$ are consistent augmented matrices in $M_{m\times(n+1)}(F)$ with the same solution set, they are row-equivalent.*
3. *For any matrix A, there is a unique RREF matrix row-equivalent to A.*

Proof.

1. Elementary row operations are reversible and preserve the validity of equations. Permuting rows changes only the order; scaling a row by $\lambda \neq 0$ creates an equivalent equation; adding λR_i to R_j creates a new equation whose solutions must still satisfy the original system.
2. If $(A|b)$ and $(A'|b')$ are consistent and have the same solutions, their respective RREF forms must also have the same solutions by (1). The RREF of a consistent system is uniquely determined by its solution set, because the parameterized solution explicitly identifies which variables are free and uniquely defines the pivot variables in terms of them. Therefore, their RREF matrices must be identical, meaning they are row-equivalent to each other.
3. Let B and B' be two RREF matrices row-equivalent to A. By (1), they have the same solution set. By (2), they must be identical since the solution set uniquely determines RREF, so $B = B'$. Therefore, there exists a unique RREF matrix that is row-equivalent to A. □

Definition 3.9 (Rank)**.** The *rank* of a matrix is the number of pivot columns in its REF or RREF.

Remark. The uniqueness of the RREF ensures that the rank of a matrix is a well-defined property, independent of the sequence of row operations used to compute it. The number of free variables is intrinsic to the solution set.

Definition 3.10 (Nullspace and Nullity)**.** For a matrix $A \in M_{m\times n}(F)$, the *nullspace* of A is the set of all solutions to the homogeneous equation $Ax = 0$:

$$\mathrm{Null}(A) := \{x \in F^n : Ax = 0\}$$

The *nullity* of A is the dimension of this nullspace.

Remark. We will rigorously define the concept of *dimension* later, but for now the *nullity* of a matrix A can be thought of as the number of free variables in the solution set of $Ax = 0$.

Theorem 3.11 (Rank-Nullity Theorem for Matrices)**.** *Let $A \in M_{m\times n}(F)$. Then:*

$$\mathit{rank}(A) + \mathit{nullity}(A) = n$$

Proof. Let R be the unique reduced row echelon form (RREF) of A. Since elementary row operations preserve the solution set of homogeneous systems, $Ax = 0 \iff Rx = 0$. Thus, $\mathrm{Null}(A) = \mathrm{Null}(R)$.

By definition, the rank of A, denoted r, is the number of pivot columns in R. Because A has exactly n columns, the variables $x_1, \ldots, x_n$ are partitioned uniquely into r pivot variables and $n - r$ free variables.

To completely describe the solution set $\mathrm{Null}(R)$, we express each of the r pivot variables strictly in terms of the $n - r$ free variables. Any solution $x \in \mathrm{Null}(A)$ is entirely and uniquely determined by assigning arbitrary values from the field F to these $n - r$ free variables.

We construct a basis for $\mathrm{Null}(A)$ by isolating the $n-r$ free variables: set one free variable to 1 and the remaining free variables to 0, and solve for the pivot variables. Doing this for each free variable generates exactly $n - r$ linearly independent vectors that span the entire solution space.

Therefore, the dimension of the nullspace (the nullity) is identically the number of free variables:

$$\mathrm{nullity}(A) = n - r$$

Rearranging this equation yields $\mathrm{rank}(A) + \mathrm{nullity}(A) = n$. □

Remark. This proof links the abstract concept of vector space dimensions to solving linear equations. The Rank-Nullity Theorem is essentially an accounting law: every column in a matrix either provides a pivot (adding 1 to the rank) or corresponds to a free variable (adding 1 to the nullity), so their sum must be n, the number of columns.

3.1.4 Structure of the Solution Space

When we solve a system $Ax = b$ using Gauss-Jordan elimination, the row operations depend only on A, not b. Consequently, if we find a solution to $Ax = b$, we are simultaneously exploring the structure of the homogeneous system $Ax = 0$.

Lemma 3.12. *Let x_p be a particular solution to $Ax = b$. Then any solution z to $Ax = b$ can be written as $z = x_p + v$, where v is a solution to the homogeneous system $Ax = 0$.*

Proof. Suppose $Az = b$ and $Ax_p = b$. By the linearity of matrix multiplication:

$$A(z - x_p) = Az - Ax_p = b - b = 0.$$

Thus, $v \coloneqq z - x_p$ is a solution to the homogeneous system $Ax = 0$. □

Remark. Geometrically, the solution set to $Ax = 0$ (the nullspace of A) forms a linear subspace (e.g., a line or plane through the origin). The solution set to $Ax = b$ is exactly this same subspace, translated away from the origin by the particular solution vector x_p. The homogeneous solutions define the ”directions” of the solution space, and x_p acts as the “anchor point”.

3.2 Matrix Inverses

3.2.1 Matrix Equations

Now we consider inverting matrices. To solve multiple systems $Ax_1 = B_1, \dots, Ax_k = B_k$ simultaneously, we do not need to apply the row reduction algorithm k times. We can solve them all at once using an augmented matrix:

$$\left(\begin{array}{c|ccc} A & B_1 & \cdots & B_k \end{array} \right) \xrightarrow{\text{Gauss-Jordan}} \left(\begin{array}{c|ccc} R & C_1 & \cdots & C_k \end{array} \right)$$

where R is in RREF and $Rx_i = C_i$ for all i.

Remark. The row operations performed during Gauss-Jordan elimination depend only on the matrix A. The target columns $B_1, \dots, B_k$ are just transformed alongside A. Since matrix multiplication acts column-wise, $R(x_1 \dots x_k) = (Rx_1 \dots Rx_k)$, the concatenated problem is strictly equivalent to the single matrix equation $AX = B$, where X and B are matrices. The j^{th} column of X is the unique solution for the j^{th} column of B.

We apply this to find the inverse of a matrix. To find a right inverse X such that $AX = I_m$, we solve the matrix equation using the augmented matrix $\left(\begin{array}{c|c} A & I_m \end{array} \right)$.

Proposition 3.13. *Let $A \in M_{m\times n}(F)$. If $AX = I_m$ has a solution, then $m \leq n$.*

Proof. Suppose $m > n$. Then A is taller than it is wide. When we perform Gauss-Jordan elimination:

$$\left(\begin{array}{c|c} A & I_m \end{array} \right) \sim \left(\begin{array}{c|c} R & C \end{array} \right)$$

Since R has n columns and is in RREF, it can have at most n pivots. Because $m > n$, R must have at least one row of zeros at the bottom. By assumption, the system $AX = I_m$ is consistent, meaning $RX = C$ is also consistent. Therefore, a zero row in R implies the corresponding row in C must also be zero (to avoid an equation of the form $0 = c \neq 0$).

However, elementary row operations are invertible. If we apply the exact same sequence of operations directly to I_m, we get C. Since I_m is already the unique RREF matrix row-equivalent to itself, and C has a zero row, we cannot row-reduce C back to I_m, contradicting that C is row-equivalent to I_m. Thus, $m \leq n$. □

Corollary 3.14. *Let $A \in M_{m\times n}(F)$. If A has a right inverse $X \in M_{n\times m}(F)$ and a left inverse $Y \in M_{n\times m}(F)$ such that $AX = I_m$ and $YA = I_n$, then A is a square matrix (i.e., $m = n$).*

Proof. Since A has a right inverse X, the equation $AX = I_m$ has a solution. By our previous proposition, this implies $m \leq n$.

Since A has a left inverse Y, we have $YA = I_n$. Taking the transpose of both sides yields $A^tY^t = I_n$. Here, $A^t \in M_{n\times m}(F)$ and $Y^t \in M_{m\times n}(F)$. Applying the same proposition to this new equation, the number of rows of A^t must be less than or equal to its number of columns, which gives $n \leq m$.

Since $m \leq n$ and $n \leq m$, we conclude $m = n$. □

Remark. The trick above relies on transposing the *left* inverse equation $YA = I_n$ to force the dimensionality bound (Proposition 3.13) in the opposite direction.

Lemma 3.15. *If A has a right inverse X and a left inverse Y, then $X = Y$.*

Proof. By associativity of matrix multiplication:

$$Y = YI_m = Y(AX) = (YA)X = I_nX = X$$

□

Corollary 3.16. *If $A \in M_{m\times n}(F)$ has both a right inverse and a left inverse, then A is square, and all left and right inverses are equal to a unique two-sided inverse, denoted A^{-1}.*

Remark. To find a right inverse X such that $AX = I_m$, row reduce $\left(\begin{array}{c|c} A & I_m \end{array} \right)$ to $\left(\begin{array}{c|c} R & C \end{array} \right)$ and solve $RX = C$. To find a left inverse Y such that $YA = I_n$, we cannot directly use row operations. Instead, we transpose the equation to $A^tY^t = I_n$. We then solve this as a right inverse problem for A^t to find Y^t, and transpose the final result to obtain Y. For a square matrix A, if a right inverse exists, it is the unique two-sided inverse A^{-1}, so we only need to perform Gauss-Jordan elimination once.

Theorem 3.17. *If $A \in M_{n\times n}(F)$ is invertible, then $\left(\begin{array}{c|c} A & I_n \end{array} \right)$ row reduces to $\left(\begin{array}{c|c} I_n & A^{-1} \end{array} \right)$.*

Proof. Since A is invertible, $AX = I_n$ has a solution. Applying Gauss-Jordan elimination yields $\left(\begin{array}{c|c} R & C \end{array} \right)$, where $RX = C$ is consistent. Since R is square, if $R \neq I_n$, it must contain a zero row at the bottom. By consistency, the corresponding row in C would also be zero. But C is row-equivalent to I_n, which is impossible if C has a zero row. Therefore, $R = I_n$.

The system becomes $I_nX = C$, meaning $X = C$. Since the inverse is unique, $C = A^{-1}$. □

Remark. We can alternatively prove Theorem 3.17 using elementary matrices. An elementary row operation on a matrix corresponds to left-multiplication by an elementary matrix E, where E is the matrix obtained from the identity by performing a single elementary row operation (exercise: prove that left multiplication by E corresponds to performing the same elementary row operation on A that was performed on E).

Now, let $A \in M_{n\times n}(F)$ be invertible, and consider the homogeneous system $Ax = 0$. Since A is invertible, we can left-multiply by A^{-1} to get $x = 0$. This means the system has only the trivial solution, which implies there are no free variables. An $n \times n$ matrix in RREF with no free variables must have a pivot in every column (by the Rank-Nullity Theorem for Matrices, Theorem 3.11), meaning its RREF is exactly I_n.

Because A row reduces to I_n, there exists a sequence of elementary row operations, corresponding to left-multiplication by elementary matrices $E_1, \ldots, E_k$, such that:

$$E_k \cdots E_1A = I_n$$

Let $E = E_k \cdots E_1$. Then $EA = I_n$. Multiplying both sides on the right by A^{-1} yields:

$$EAA^{-1} = I_nA^{-1} \implies E = A^{-1}$$

Applying this identical sequence of row operations to the augmented matrix $\left(\begin{array}{c|c} A & I_n \end{array} \right)$ corresponds to left-multiplying the entire block matrix by E:

$$E \left(\begin{array}{c|c} A & I_n \end{array} \right) = \left(\begin{array}{c|c} EA & EI_n \end{array} \right) = \left(\begin{array}{c|c} I_n & E \end{array} \right)$$

Since we established $E = A^{-1}$, the augmented matrix reduces exactly to $\left(\begin{array}{c|c} I_n & A^{-1} \end{array} \right)$.

3.2.2 Preview of Determinants

When is a 2×2 matrix invertible? From the previous theorem, A is invertible if and only if its RREF is I_2.

Proposition 3.18. *A 2×2 matrix A is invertible if and only if its rows R_1 and R_2 are not scalar multiples of each other.*

Proof. ($\Longrightarrow$) If R_1 and R_2 are multiples of each other, row reduction will produce a row of zeros. Thus, the RREF is not I_2, and A is not invertible.

($\Longleftarrow$) If the rows are not multiples, then the first column cannot be entirely zero (otherwise R_1 and R_2 would be multiples of each other). After a possible row swap, we have:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \sim \begin{pmatrix} 1 & b' \\ 0 & d' \end{pmatrix}$$

where $a \neq 0$. Since R_2 is not a multiple of R_1, the row reduction $R_2 \to R_2 - \frac{c}{a} R_1$ leaves a non-zero entry d'. Thus, the second column contains a pivot, making the RREF I_2. $\square$

Remark. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the condition that the rows are not multiples means $ad \neq bc$. We define the determinant as $\det A := ad - bc$. Thus, A is invertible if and only if $\det A \neq 0$.